

WEEK-11

TASK-1: To Connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reliability, Refactor 4 Components

```
root@LAPTOP-RJFDBEST:~# nano input.h
root@LAPTOP-RJFDBEST:~# nano input.c
root@LAPTOP-RJFDBEST:~# nano process.h
root@LAPTOP-RJFDBEST:~# nano process.c
root@LAPTOP-RJFDBEST:~# nano validate.h
root@LAPTOP-RJFDBEST:~# nano validate.c
root@LAPTOP-RJFDBEST:~# nano output.h
root@LAPTOP-RJFDBEST:~# nano output.c
root@LAPTOP-RJFDBEST:~# nano main.c
root@LAPTOP-RJFDBEST:~# ls -l
total 164d
-rw-r--r-- 1 root root 135 Aug 12 06:53 File1.txt
-rw-r--r-- 1 root root 64 Aug 8 08:42 "File1.txt"
-rw-r--r-- 1 root root 155 Aug 12 06:58 File2.txt
drwxr-xr-x 11 root root 4096 Aug 23 18:09 OSCP
drwxr-xr-x 2 root root 4096 Aug 20 05:42 SKILL_Week_4
-rw-r--r-- 1 root root 20128 Sep 18 09:11 append_redirect
-rw-r--r-- 1 root root 2918 Sep 18 09:11 append_redirect.c
-rw-r--r-- 1 root root 21128 Sep 18 11:51 background_jobs
-rw-r--r-- 1 root root 3288 Sep 18 11:51 background_jobs.c
-rw-r--r-- 1 root root 21352 Sep 18 04:07 builtin_shell
-rw-r--r-- 1 root root 3613 Sep 18 04:07 builtin_shell.c
-rw-r--r-- 1 root root 7433 Sep 18 03:52 builtin_shell.c.save
-rw-r--r-- 1 root root 12508 Sep 18 04:08 cd_navigation.c
-rw-r--r-- 1 root root 3683 Sep 18 04:08 cd_navigation.c
-rw-r--r-- 1 root root 16464 Sep 16 18:51 client
-rw-r--r-- 1 root root 2123 Sep 16 18:58 client.c
-rw-r--r-- 1 root root 73 Sep 18 09:44 combined.txt
-rw-r--r-- 1 root root 20176 Sep 18 09:31 combined_redirect
-rw-r--r-- 1 root root 2920 Sep 18 09:38 combined_redirect.c
-rw-r--r-- 1 root root 16576 Aug 17 05:59 command_executor
-rw-r--r-- 1 root root 1152 Aug 17 05:59 command_executor.c
-rw-r--r-- 1 root root 16208 Sep 18 04:55 command_history
-rw-r--r-- 1 root root 3589 Sep 18 04:55 command_history.c
-rw-r--r-- 1 root root 26072 Sep 18 18:17 complex_command
-rw-r--r-- 1 root root 3988 Sep 18 10:17 complex_command.c
-rw-r--r-- 1 root root 16216 Aug 29 10:07 copy
-rw-r--r-- 1 root root 727 Aug 29 09:48 copy.c
-rw-r--r-- 1 root root 727 Aug 29 10:07 'copy.c '
-rw-r--r-- 1 root root 358 Sep 27 13:25 copy_low.txt
-rw-r--r-- 1 root root 358 Sep 27 13:25 copy_low.txt
-rw-r--r-- 1 root root 16184 Aug 29 09:52 copyfile
-rw-r--r-- 1 root root 904 Aug 29 09:51 copyfile.c
-rw-r--r-- 1 root root 16048 Sep 17 17:09 copy_low.txt
-rw-r--r-- 1 root root 1493 Sep 17 17:09 cow_demo.c
-rw-r--r-- 1 root root 40 Sep 17 17:09 cow_demo.c
-rw-r--r-- 1 root root 16528 Sep 27 15:24 deadlock
-rw-r--r-- 1 root root 1582 Sep 27 15:24 deadlock.c
-rw-r--r-- 1 root root 16048 Sep 27 15:22 deadlock_prevention
-rw-r--r-- 1 root root 1728 Sep 27 15:22 deadlock_prevention.c
-rw-r--r-- 1 root root 40 Sep 17 18:18 destination.txt
-rw-r--r-- 1 root root 1706 Sep 17 17:41 dynamic_memory
-rw-r--r-- 1 root root 1490 Sep 17 17:41 dynamic_memory.c
-rw-r--r-- 1 root root 60 Sep 18 09:23 error.txt
-rw-r--r-- 1 root root 16508 Sep 27 13:25 filecopy
-rw-r--r-- 1 root root 3335 Sep 27 13:23 filecopy.c
-rw-r--r-- 1 root root 16884 Sep 27 16:07 foreground_switch
-rw-r--r-- 1 root root 3986 Sep 27 16:06 foreground_switch.c
-rw-r--r-- 1 root root 16304 Aug 18 03:51 fork_process
-rw-r--r-- 1 root root 1179 Aug 18 03:58 fork_process.c
-rw-r--r-- 1 root root 53 Aug 29 08:13 information.txt
-rw-r--r-- 1 root root 4096 Sep 27 15:19 inode_test
-rw-r--r-- 1 root root 249 Sep 27 17:08 input.c
-rw-r--r-- 1 root root 59 Sep 27 17:08 input.h
-rw-r--r-- 1 root root 71 Sep 27 13:35 input.txt
-rw-r--r-- 1 root root 16048 Sep 18 08:38 input_redirect
-rw-r--r-- 1 root root 1578 Sep 18 08:38 input_redirect.c
-rw-r--r-- 1 root root 16812 Sep 27 16:42 interrupt_handling
-rw-r--r-- 1 root root 3983 Sep 27 16:41 interrupt_handling.c
-rw-r--r-- 1 root root 16864 Sep 27 16:28 job_recovery
-rw-r--r-- 1 root root 5762 Sep 27 16:28 job_recovery.c
-rw-r--r-- 1 root root 21912 Sep 18 11:59 job_table
-rw-r--r-- 1 root root 5236 Sep 18 11:59 job_table.c
-rw-r--r-- 1 root root 16624 Sep 27 15:58 jobs
-rw-r--r-- 1 root root 3460 Sep 27 15:53 jobs.c
-rw-r--r-- 1 root root 5825 Sep 18 09:04 listing.txt
-rw-r--r-- 1 root root 16302 Sep 16 08:08 ls_grep
-rw-r--r-- 1 root root 1367 Sep 16 08:18 ls_grep.c
-rw-r--r-- 1 root root 863 Sep 27 17:12 main.c
-rw-r--r-- 1 root root 16488 Sep 17 16:25 memory_layout
-rw-r--r-- 1 root root 1965 Sep 17 16:25 memory_layout.c
-rw-r--r-- 1 root root 19768 Sep 18 08:28 multi_pipe
-rw-r--r-- 1 root root 5843 Sep 18 08:28 multi_pipe.c
-rw-r--r-- 1 root root 11 Sep 18 09:01 normal.txt
-rw-r--r-- 1 root root 70 Sep 18 09:44 order.txt
-rw-r--r-- 1 root root 488 Sep 27 17:12 output.c
-rw-r--r-- 1 root root 129 Sep 27 17:11 output.h
-rw-r--r-- 1 root root 18 Sep 27 13:36 output.txt
-rw-r--r-- 1 root root 20888 Sep 18 09:01 output_redirect
-rw-r--r-- 1 root root 2719 Sep 18 08:58 output_redirect.c
-rw-r--r-- 1 root root 19304 Sep 17 19:09 path_resolver
-rw-r--r-- 1 root root 2292 Sep 17 19:08 path_resolver.c
-rw-r--r-- 1 root root 18368 Sep 18 08:04 pipe_dataflow
-rw-r--r-- 1 root root 2488 Sep 18 08:04 pipe_dataflow.c
-rw-r--r-- 1 root root 22224 Sep 18 05:05 pipeline
-rw-r--r-- 1 root root 6196 Sep 18 05:05 pipeline.c
-rw-r--r-- 1 root root 16216 Aug 19 18:44 process
-rw-r--r-- 1 root root 173 Sep 27 17:18 process.c
-rw-r--r-- 1 root root 40 Sep 27 17:09 process.h
-rw-r--r-- 1 root root 850 Aug 19 18:21 process_demo.c
-rw-r--r-- 1 root root 16204 Aug 18 04:12 process_exp
-rw-r--r-- 1 root root 973 Aug 18 04:11 process_exp.c
-rw-r--r-- 1 root root 16680 Sep 27 17:03 process_group_control
-rw-r--r-- 1 root root 3902 Sep 27 17:02 process_group_control.c
-rw-r--r-- 1 root root 16512 Aug 19 18:25 process_states
-rw-r--r-- 1 root root 1685 Aug 19 18:24 process_states.c
-rw-r--r-- 1 root root 6 Aug 18 04:18 process_states.c.save
-rw-r--r-- 1 root root 16408 Sep 16 06:04 producer_consumer
-rw-r--r-- 1 root root 1899 Sep 16 06:53 producer_consumer.c

root@LAPTOP-RJFDBEST:~# nano validate.h
root@LAPTOP-RJFDBEST:~# nano process.c
root@LAPTOP-RJFDBEST:~# cat validate.h
#define VALIDATE_H
#define VALIDATE_H

int validate_number(int number);
int validate_result(int result);

#endif

root@LAPTOP-RJFDBEST:~# cat process.c
#include "process.h"

int calculate_square(int number)
{
    return number * number;
}

root@LAPTOP-RJFDBEST:~# gcc -Wall -Wextra main.c input.c process.c validate.c output.c -o modular_system
root@LAPTOP-RJFDBEST:~# ./modular_system
Modular End-to-End System
=====
Enter a positive integer: 5
=====
End-to-End Execution Result
=====
Input : 5
Square : 25
STATUS : SUCCESS
=====
root@LAPTOP-RJFDBEST:~#
```

TASK-2: TO Detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios

```
root@LAPTOP-RJFDBEST:~# nano error_handling.c
root@LAPTOP-RJFDBEST:~# ls -l ./error_handling.c
-rw-r--r-- 1 root root 1828 Sep 27 07:18 error_handling.c
root@LAPTOP-RJFDBEST:~# gcc -Wall -Wextra error_handling.c -o error_handling
root@LAPTOP-RJFDBEST:~# ./error_handling
error: ./error_handling: cannot open shared object file: No such file or directory
root@LAPTOP-RJFDBEST:~# ./error_handling
error: ./error_handling: cannot open shared object file: No such file or directory
root@LAPTOP-RJFDBEST:~# ./error_handling
error: ./error_handling: cannot open shared object file: No such file or directory
=====
1. Test Invalid Input
2. Test Division By Zero
3. Test File Error
4. Display Error Log
5. Exit
Enter choice : 1
Enter an integer: abc
18000: invalid syntax: input is not an integer.
1. Test Invalid Input
2. Test Division By Zero
3. Test File Error
4. Display Error Log
5. Exit
Enter choice : 1
Enter an integer: 123abc
18000: invalid syntax: unexpected characters found.
1. Test Invalid Input
2. Test Division By Zero
3. Test File Error
4. Display Error Log
5. Exit
Enter choice : 1
Enter an integer: 25
Valid integer received: 25
1. Test Invalid Input
2. Test Division By Zero
3. Test File Error
4. Display Error Log
5. Exit
Enter choice : 1
Enter an integer: 18
Valid integer received: 18
1. Test Invalid Input
2. Test Division By Zero
3. Test File Error
4. Display Error Log
5. Exit
Enter choice : 5
Exiting gracefully.
root@LAPTOP-RJFDBEST:~#
```

TASK-3: To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings

[illegible]